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Chapter 3

Production Schedule Control



Production Schedule Control

Introduction

Scheduled production is a term with as many definitions as companies to which it is applied. In fact most organisations involve some activities of a schedule nature. And conversely, most organisations involve some activities which are best controlled by production orders. Schedule characteristics include, but are not limited to:

- Production which is driven by daily schedules
- Production which is balanced between work stations
- Product sequencing which is efficient
- Product variations which are limited
- Product structures which are shallow
- Production runs which are high volume
- Capacity which is dedicated
- Tooling which is specialised
- Routes which are fixed
- Processing time which is short
- Set up/change-overs which are rapid
- Work queues which are small and planned
- Transaction reporting which is minimal
- Systems which are paperless

Clearly, most of these characteristics describe not any inherent aspect of product or process, but rather the degree to which specific types of solution have been implemented. These solutions tend to employ techniques such as:

- Linking to Customer Schedules
- The elimination of product orders because schedules produce no paperwork
- Production route and structure definitions
- The use of clear, simple and appropriate production signals
- WIP inventory used to net gross requirements
- Automatic backflushing of materials and operations
- Interactive status enquiry and reporting capabilities
- Enhanced costing and reporting systems
- Bucketless/net change production planning

This type of environment is often referred to as repetitive production environment. The use of the term repetitive implies the use of schedules to control production rather than production orders.

Schedule Control Activities

The Schedule Control function has been designed as a flexible solution capable of fulfilling the needs of manufacturers operating in widely disparate repetitive environments.

The recurring activities which comprise management of the schedule functions fall into five functional areas:

Requirements Planning

Functions that determine which items are required to be manufactured and purchased, in what quantity, and by what date. See Materials Requirements Planning.

Figure 1: Materials Requirements Planning

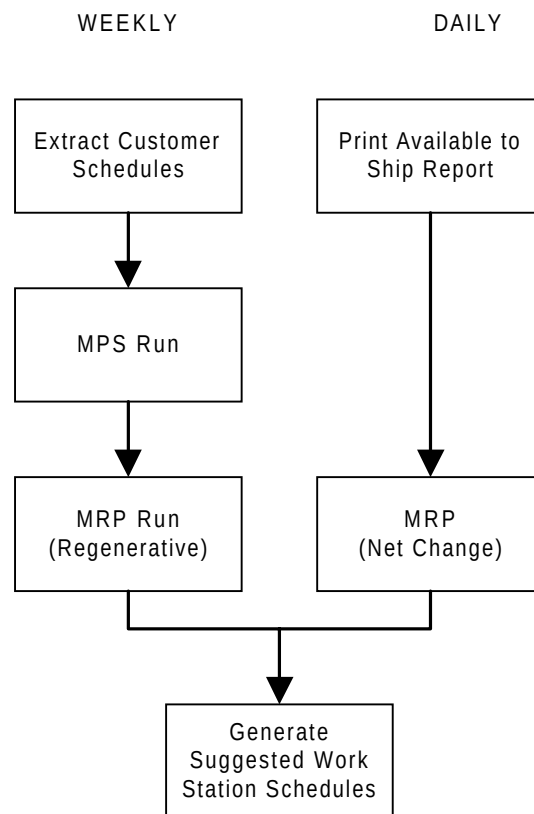


Figure 2: Work Station Scheduling & Review

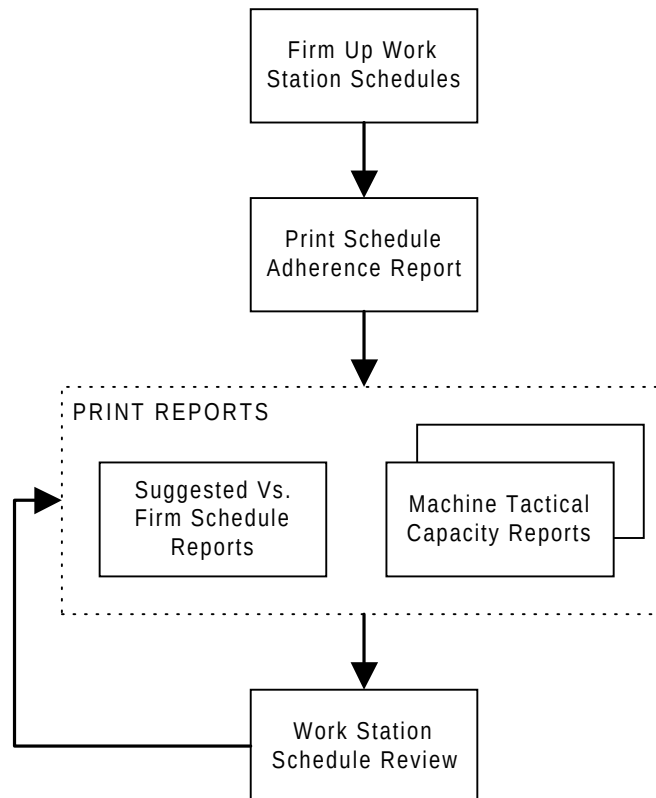
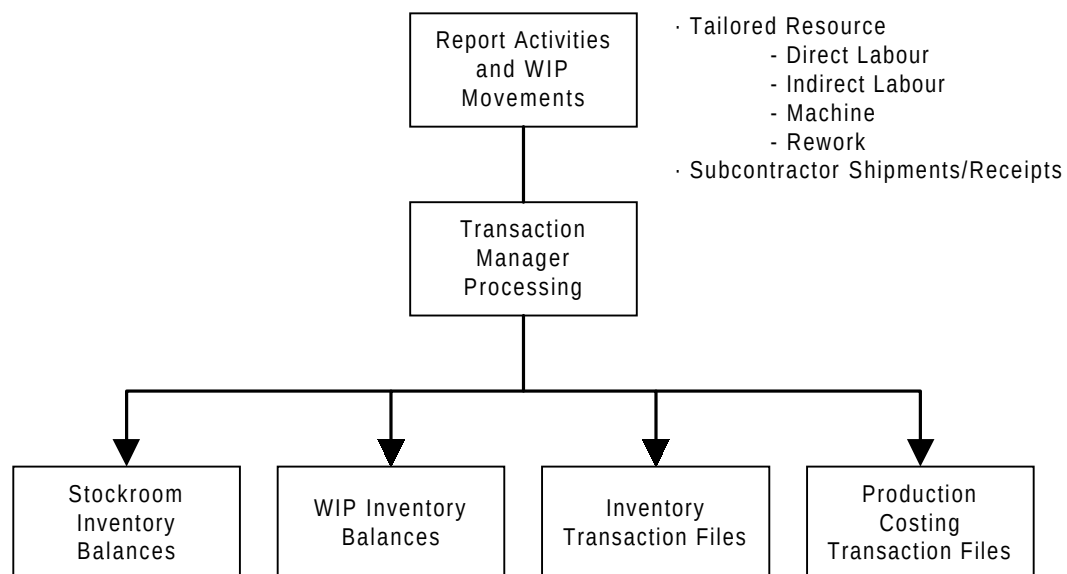


Figure 3: Production And WIP Reporting



Work Station Scheduling and Review

Functions supporting firming, reviewing and manipulating work station schedules.

Work Station Control

Functions which communicate the action plan to work station and other shop floor resources.

Production/WIP Reporting

Functions which allow resource activity to be reported from the shop floor, including labour, machine and WIP inventory activities.

Measuring Performance

Functions which provide performance measures with which to evaluate efficiency, schedule adherence, material usage, scrap, WIP valuation etc.

Requirements

The use of the planning functions MPS and MRP to generate the suggested schedules is covered in detail in Production Planning.

The purpose of this course is to understand how schedules may be controlled and manipulated after they have been created.

However, a brief summary of the way in which the demand for items is established is necessary before a proper understanding of schedule control can be achieved.

Establishing Demand

Demand for any schedule controlled items comes from the MPS/MRP functions. It consists of a combination of:

- Customer order demand
- Sales forecasts demand
- Manual requirements
- Demand generated from planned parent item replenishment

If forecast demand is used there are a number of alternatives:

- Forecasts created within the MPS and MRP functions themselves
- Forecasts created using the Forecasting application and subsequently transferred to MPS and MRP
- Extraction of customer schedules from the Advanced Customer Scheduling (AC) application and the building of aggregated item forecasts from them

MPS/MRP Planning for Schedules

Schedule functions are supported by four distinct features within MPS and MRP. These include:

- Schedules may be planned for an item rather than production orders
- WIP inventory used to net gross requirements
- Overlapping operations and back scheduling
- Generation of suggested operation work schedules

Planning Replenishments as Suggested Schedules

- Schedules produce no paperwork
- Schedules require no reporting
- Schedules may be fully dynamic or firmed in batch on the basis of user data range

- Schedules planned in MPS/MRP are the basis for daily work station operation schedules
- Schedule quantities could be used to create production orders (on an exception basis only)

The Use of Work in Process Inventory in Calculating Net Requirements

In the absence of production orders there is a level of work-in-progress inventory which needs to be recognised by the planning system in order to avoid over-planning materials and production.

WIP inventory balances are available at operations defined as count points on the planning route for an item, and are taken into consideration in establishing the net demand in the planning routines.



Note: Queue time is only effective on the second and subsequent operations.



Note: Queue time is not shown on the schedule review BUT is allowed for in the scheduling calculations.



Note: Start dates are calculated backwards from the due date.



Note: Operation schedule is calculated forwards from the start date.



Note: Move time is treated in the same way as queue time.

Normal Sequence Of Events

Schedule (Weekly)

Extract Customer Schedules

Run MPS

Review MPS

Change MPS Flow Line Item(s) (if required)

Run MRP

Review MRP

Create Work Station Schedule

Review Work Station Load or Manage Work Station Load

Confirm Work Station Schedule

Reschedule (Daily)

Run Net Change MRP (see Run MRP)

Create Work Station Schedules (see Confirm Work Station Schedule)

Review Work Station Load or Manage Work Station Load

Reschedule Flow Line Item (if required) (see Manage Work Station Load)

Work Station Schedule Review

When the schedules are first created they have a status of Suggested. They are shown against each of the work stations defined in the route.

Work station schedule review provides an effective tool for productively assessing and appropriately manipulating schedules so that they are consistent with actual production conditions and input availability.

Firming Schedules

In a typical implementation, following refresh of suggested daily work station schedules, a procedure is run to firm these schedules up to some future date. This firming procedure may be performed at any interval, though it is most likely performed daily, weekly, or somewhere in between.

The frequency with which this procedure is performed relates very specifically to the environment in which the schedule application is used.

The Firm Period, that period in which schedules have been firmed, is a very significant time interval within the application. For this reason, substantial consideration should be given to the frequency, timing and period interval used in firming schedules. Suggested schedules which are created from the planning applications do not replace or refresh firmed work station schedules.

Firm work station schedules are a major advantage for many organisations, allowing manufacturing a stable definition of the production plan for some period, however brief, into the future.

On the other hand, this technique stops production from being affected by production planning. For companies producing exclusively to customer order and for whom the customer orders are dynamic, segregating actual requirements from work station schedules may be undesirable.

Most companies find there is a reasonable compromise which allows some schedule firming for productivity and control without threatening customer service.

This is especially true using scheduling because it includes features which identify variations between suggested and firm item schedules and projected MPS inventory variances form safety stock objectives.



Note: The action of firming work station schedules may (optionally) cause the supply status of the item in MPS/MRP to be changed to CS - Confirm Schedule. Remember that if MPS/MRP is rerun the CS items is NOT deleted but replanned - according to the rescheduling (filter) settings active for the confirmed schedules type of supply.

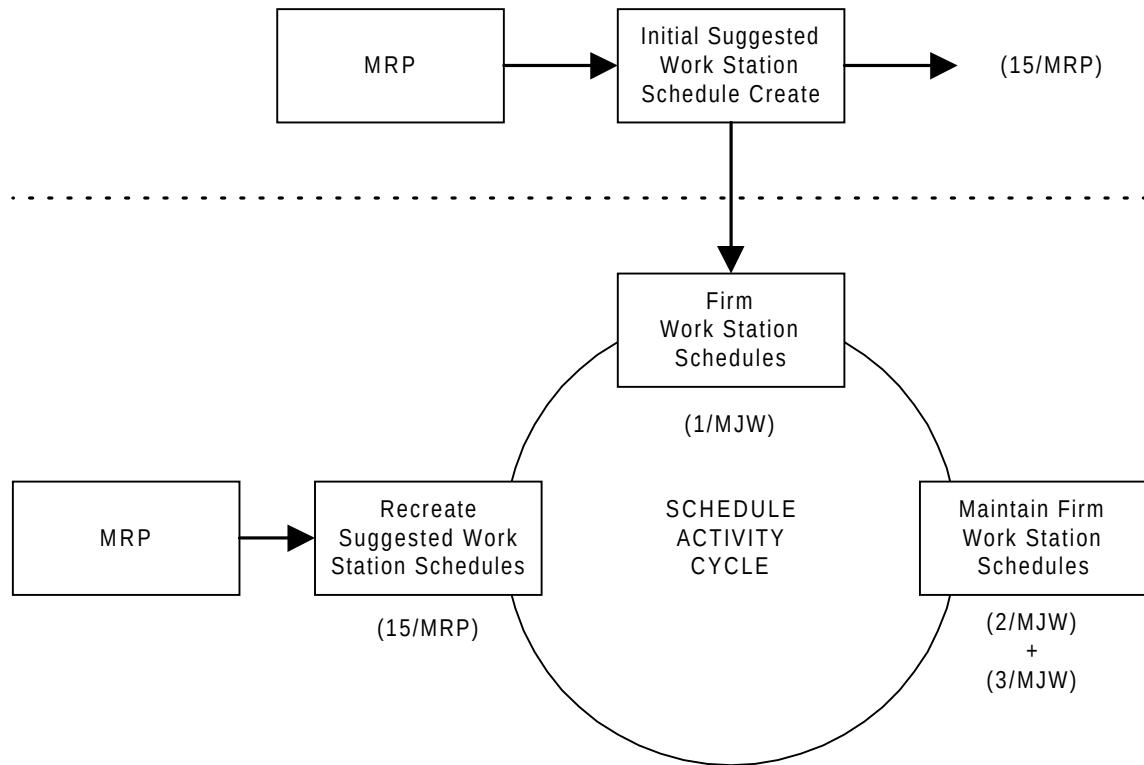
Maintaining Schedules

Only firm work station schedules are maintainable. That is, so long as a schedule has not been firmed, it still belongs to the planning system and may be replaced anytime the planning system is used to recreate suggested work station schedules.

Firm work station schedules, however, always belong to the user and is not updated by the system (though the system still plans suggestions and make them available for comparison with firmed schedules).

Another consideration relating to the Firm Period is that a number of important functions differentiate between schedules which are firm and those which are not. The Daily Material Pull report, for example, includes only those materials required to support firm schedules.

Schedule Activity Cycle



Lot Tracking

Lot traceability is supported by the use of a system generated WIP lot number in place of the work order number as used in discrete production. The WIP lot number is linked to the item and the route. They uniquely identify transactions that are created by scheduled issues and receipts. Inventory Management contains enquiries and reports for tracking the origins of items used within the production system.

Generate Warehouse Requirements

1...
2... **1/MKS**

The generation of warehouse requirements for a schedule is achieved through a two stage process:

- Daily Material Pull List
- Generate Warehouse Requirements

The first of these submits a batch job, while the second is almost identical to the Generate Warehouse Requirements option for production orders.



Select Generate Warehouse Requirements on the Production Schedule Control menu. Various selection parameters are available.

This option identifies control numbers for schedules. These are used by the Warehouse application to control movements within that application. This technique stops production from being affected by Production Planning.

Following the running of the Daily Material Pull List and Generate Warehouse Requirements the Picking/Confirm Picking and Material Issues to Production options in the Warehousing application must be completed before booking of materials/operations can occur in Production Control.

Issue Material

11/MKS

This option is used to issue (confirm) Material Policy Code 3 items to the shop floor stockroom. These are then reserved and must be formally consumed at the operation in which they are used.



When Issue Material is selected on the Production Schedule Control menu, the Material Issue screen is displayed.

MK222	Z1 - GB Demonstration Company	USER GBJBAMS	8/12/95 17:14:56
Material Issue			
Item.....?	_____		
Stockroom.....?	_____		
Lot.....?	_____		
Quantity.....	_____	.000	
Parent Item.....?	_____		
Route.....?	_____		
Operation.....?	_____		
Input Sequence.....?	_____		
Work Station.....?	_____		
Reference.....	_____		
Transaction Date.....	8/12/95		
Transaction Type.....?	_____		
Reason.....?	_____		
Reversal.....	0	(0/1)	
F3=Exit F4=Prompt F8=Update F15=View Floor Stock			
05-31  MW KS IM II KENB20A PET166S2			

Fields:

Item

Enter the item which is to be issued to floor stock. This must be a valid item with a Material Control Policy flag setting of 3 (issue to floor stock).

Stockroom

Enter the stockroom from which the item is to be issued. This is a mandatory field and must be a valid stockroom to which the item is defined.

Lot

If the item is lot controlled, enter the lot from which the quantity is being issued.

Quantity

Enter the quantity which is to be moved from the Inventory issuing stockroom to the shop floor stockroom.

Parent Item

Enter the product/parent item on or in which the input is to be used. This is a mandatory field.

Route

This field is used to specify the issue of materials to flow routes. A route may not be specified for process groups or non flow, single output routes.

If the relevant route is the planning (discrete) route, then a code must not be entered.

If the relevant route is a flow route, then the route code may be entered (unless it is blank).

Operation

Enter the operation sequence in the route for the parent item in which the issuing item is used as an input. This is a mandatory field.

Input Sequence

This is a mandatory field which is four characters in length. If the quantity entered is greater than that required, a warning message is displayed. The excess quantity is not allocated but becomes free stock in the floor stock location.

Work Station

Enter the work station where the item is to be used. Entry is mandatory. The work station must have a valid floor stock location assigned to it through the Maintain Work Station/WIP Location facility, before the material issue is permitted. If a valid code has been specified in the Route field, then the standard work station on the flow route must be entered.

Reference

Enter a reference for this issue.

This allows an inventory lot to be linked to a production reference therefore, where applicable, to a production lot number. This facilitates the tracking of material through the production route.

Transaction Date

Enter the date the material is required, as displayed on the Daily Material Pull List.

Transaction Type

The issue is assigned to a transaction type. A transaction type is used for inventory analysis; it does not effect changes to item balances. Entry is mandatory.

Transaction types are defined in the Inventory Management application, descriptions file against type TRAN.

Reason

Enter a reason code explaining the material issue. Entry is mandatory.

This must be a valid reason code set up on the Inventory Management descriptions file against major type MOVR.

Reversal

If materials which have been issued to the floor need to be returned to an Inventory stockroom, this flag can be set to 1 and an issue reversal can be achieved.



Note: In this circumstance, the stockroom should be the receiving stockroom, the quantity should be the amount that is being returned from the shop floor, and the parent item, operation and work station should indicate the floor location the item was originally issued to.

0 - (The default) to allow standard issue of items from Inventory to Production.

1 - To indicate material is being returned to an Inventory stockroom from a shop floor stockroom.

Functions:

F15

View Floor Stock. Available after entering entry fields. Displays the items in shop floor stock which are available for issue.

Item Booking



12/MKS



When Item Booking is selected on the Production Schedule Control menu, the Item Booking - Header screen is displayed.



Note: For items with no inspection order but with an inspection policy of 2 (Bookings), an inspection order is automatically generated for each schedule booking made at a quality controlled operation.

Transactions

Production Schedule bookings provide facilities to make bookings against WIP for items subject to schedule control.



Note: Bookings are not made to a schedule but for the WIP associated with the operation.

As the WIP is booked this provides a supply to meet the schedule demand. When rescheduling takes place if there is sufficient supply then a new schedule is NOT suggested. Suggested schedules are only produced when there is insufficient supply of WIP to meet the demand, that is, when some product is made.

As with Production Order bookings the activities you wish to report must have been defined using the Maintain Activity Control File function. These definitions also determine the information to be included in transaction files and the production activity database records to be written. This in turn determines the information which can be extracted by AFI for posting to the General Ledger application.

Bookings can be made in two ways:

Item Bookings - This option allows an operator or crew to book a number of activities they have performed without the need to re-enter information about themselves for every event reported. The default route for standards calculations is the planning route associated with the item booked.

Operator Bookings - This option allows the entry of heading information about an item without the need to input this again for every booking. A route code may be entered for flow route items only. The default route for standards calculations is the planning route associated with the item booked.

The Item and Operator booking options each utilise the same processing mechanism for doing all types of WIP bookings. Activity can be reported through either method for any activity types that have been established for the company.

These facilities are very similar to those provided to book production orders.

Backflushing Time and Materials Summary

	Op. 10	OP 20
	Backflushed	Count Point

Production Order		
WORDRP (Time)	Not Allowed	Allowed
Formal Issue (Material)	Manual issue allowed	Manual issue allowed
Backflushed (Material)	Should not be allowed	Issued for Op 10 & 20 on booking completed operation.
OTHERXX (Time)	Not Allowed	Allowed
Scheduled Item		
PRODRP (Time)	Not Allowed if a WIP movement	Allowed
Formal Issue (Material)	Not Allowed if a WIP movement	Issued as bookings are made.
Backflushed (Material)	Not Allowed if a WIP movement	Issued as bookings are made.
OTHERXX (Time)	Not Allowed if a WIP movement	Allowed

Item Booking Header Screen



When Item Booking is selected on the Production Schedule Control menu, the Item Booking Header screen is displayed.

MK200	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:07:58
Item Booking			
Header details			
Date.....	8/12/95	Shift	__
Work Station.?	__	Dept.?	__
Operator.....?	__	Cr.?	__
F3=Exit F4=Prompt			
05-18	82	MW	KS IM II KENB20A PET166S2

Fields:

Date

Enter the date for this booking. This field defaults to the current date, but can be amended. All bookings made during this session are recorded against the date entered.

Shift

Enter the shift number to which the booking relates. This provides a more precise time frame for recording booking if a start and finish time are not entered. For example, if the booking has an elapsed time of 3 hours then this might be linked to shift 2.

If entered, this should be a shift number on the shift profile in force on the booking date at the work station at which the booking was made.

Work Station

Enter the work station against which the booking is to be made. This would usually be the work station assigned to the operation on the route.



Note: If the planning route is a flow route, then entry of the associated standard work station is mandatory.

If a non standard work station (that is, not the work station expected by the route) is entered, any WIP inventory resulting from the booking is allocated to the WIP

location associated with the nominated work station; which may differ to the one on the route.

If the WIP location differs, the system displays a window stating Reporting WIP location is not the standard for this operation and continuation must be manually confirmed.

If the operation involves utilisation of floor stock items, then any non standard work station must be associated with the same floor stock location in order to issue materials. If it is not, a materials transfer must be actioned to the new location.

A work station and/or department must be recorded against each booking.

Department

Departments are an analysis tool and do not effect any constraints or control over WIP inventory, work station, operator or crew bookings.

For each booking, a work station must be defined.

If a work station is entered, but not a department, the system defaults to and displays any department assigned to the work station definition when Enter is pressed.

If a work station is not entered, but a department is, bookings details assumes the standard work station defined for the operation on the route.

If a department is not entered, but the operator has been assigned a department via Operator Maintenance, the system defaults to and displays that department when Enter is pressed.

Departments are user-defined and are entered on the system via the Maintain Parameters File facility, against type DEPT.

Operator

Enter an operator code identifying the person who performed the work.

Alternatively, a crew code may be entered if a crew (group of operators) rather than an individual was responsible for the work being reported.

Operator and crew codes are mutually exclusive.

Cr

Crew. Enter a crew code identifying the people who performed the work.

Alternatively, an operator code may be entered if an individual, rather than a crew was responsible for the work. Crew and operator codes are mutually exclusive.

Any booked item associated with the operator is allocated to each member of the crew automatically. For example, if an operator time of 8 hours is entered, each member of the crew is assumed to have worked 8 hours on the operation. To amend these times, use F13=Crew Details, which allows you to modify the time on an individual basis.



When **Enter** is pressed on the Item Booking - Header screen, the Item Booking - Details screen is displayed.



Select **F8** to end session and update the transaction files.

Item Booking Details Screen

 When **Enter** is pressed on the Item Booking Header screen, the Item Booking Details screen is displayed.

This screen allows the item information to be entered. The fields are the same as those for Production Order Bookings. The only exceptions are that an Item Number is required instead of an order number and that the completion options are omitted.

The Item Booking - Details screen contains the following fields:

 *Note: If Operator and Crew fields are left blank but operator times are booked the labour hourly rates from the cost centre associated with the quoted work station are used to cost the labour booking.*

 *When all required fields are completed and components/outputs confirmed as required, the booking details window is displayed and updated after each booking entry in the session.*

MK200	Z1 - GB Demonstration Company	User GJBAMS	8/12/95 17:07:58
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Item Booking	
Header details	
Date.....	8/12/95 Shift __
Work Station.?	WS01 Dept.? __
Operator.....?	Cr.? __
Booking Details	
Activity.....?	PRODRP Rev Ind 0
Item.....?	
Operation.....?	Alt en? __
Lot Number....	
Time	
Oper Elapsed.	.00
W/S Elapsed.	.00
Quantity	
Good..	.000
Scrap.	.000 Reason.? SC
Held..	.000 Reason.? QI

F3=Exit	F4=Prompt	F12=Previous	F13=Crew Details
F14=Component Details	F16=Outputs	F17=Scrap	F18=Hold
10-10	SP	MW	KS
		IM	II
		KENB20A	PET166S2

Functions:

F13

Crew Details. Display Crew Details window.

F14

Component Details. Display Component Details window.

F16

Outputs. Display Outputs window.

F17

Scrap. Display Scrap WIP Details window.

F18

Hold. Display Held WIP Details window.

F19

Booking Details. Display Booking Details window.

Operator Booking



Note: For items with no inspection order but with an inspection policy of 2 (Bookings), an inspection order is automatically generated for each operator booking made at a quality controlled operation.



13/MKS



When Operator Booking is selected on the Production Schedule Control menu, the Operator Booking Header screen is displayed.

Operator Booking Header Screen



When Operator Booking is selected on the Production Schedule Control menu, the Operator Booking Header screen is displayed.

The Operator Booking - Header screen contains the following fields:

MK200	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:10:39
Operator Booking			
Header details			
Date.....	8/12/95	Shift	__
Work Station.?	__	Dept.?	__
Item.....?	__	R?	__
Operation....?	__	Altern.?	__
Lot Number....	__		
F3=Exit F4=Prompt			
05-10	82	MW	KS IM II KENB20A PET166S2

Fields:

The work station must be entered. Other fields are validated according to the Activity Type being reported on the next screen.

R

Route. This field enables entry of an alternative predefined flow route in preference to the standard planning route.

The system verifies that the work station entered is the standard one defined on the route/operation.



When **Enter** is pressed on the Operator Booking - Header screen, the Operator Booking - Details screen is displayed.

Operator Booking Details Screen



When **Enter** is pressed on the Operator Booking Header screen, the Operator Booking Details screen is displayed.

This screen allows the operator information to be entered. The other fields are the same as the previous option.

MK200	Z1 - GB Demonstration Company	User GJBAMS	8/12/95 17:10:39
Operator Booking			
Header details			
Date.....	8/12/95	Shift	__
Work Station.?	WS01	Dept.?	__
Item.....?	MS002	R?	__
Operation....?	10	Altern.?	__
Lot Number....	__		
Booking Details			
Activity.....?	PRODRP	Rev Ind	0
Operator.....?	█	Cr.?	__
Time	__		
Oper Elapsed.	.00		
W/S Elapsed.	.00		
Quantity			
Good..	.000		
Scrap.	.000	Reason.?	SC
Held..	.000	Reason.?	QI
F3=Exit F4=Prompt F12=Previous F13=Crew Details F14=Component Details F16=Outputs F17=Scrap F18=Hold F19=Booking Details 12-10 S MW KS IM II KENB20A PET166S2			

When all required fields are completed and components/outputs confirmed as required, the booking details window is displayed and updated after each booking entry in the session.

Subcontract WIP Shipper



14/MKS



When Subcontractor WIP Shipper is selected on the Production Schedule Control menu, the Subcontractor WIP Shipper Entry screen is displayed.

This option performs the same task as the Subcontractor WIP Shipper option on the Production Order Control menu but of course for schedule controlled items. An item number is required instead of an order number.

Subcontract Material Shipper



15/MKS



Select Subcontract Material Shipper on the menu to display the Subcontractor Material Shipper/Return screen.

This option performs the same tasks as the Subcontractor Material Shipper option on the Production Order Control menu but of course for schedule controlled items.

Receive from Subcontractor



16/MKS



When Receive From Subcontractor is selected on the Production Schedule Control menu, the Subcontractor Order Receipts Entry screen is displayed.

This option performs the same tasks as the Receive From Subcontractor option on the Production Order Control menu but of course for schedule controlled items.

An item number is required instead of an order number.

Item Production Schedule Report

1...
2...

21/MKS

This report shows the scheduled production for an item within a selected date range. The scheduled quantities are summarised in time slots which reflect the start date and reporting profile (period) definitions for the related MPS model. These buckets might reflect days for the short term and weeks for future dates. The report presents up to eight columns, the last one includes all production schedules from the end of the previous bucket up to the end date selected.

The report also provides a time slot selecting the current WIP inventory position.



When Item Production Schedule is selected on the Production Schedule Control menu, the Item Production Schedule Report Selection screen is displayed.

MJ650	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:15:25
Item Production Schedule Report Selection			
Organisational Model...? __			
WIP Locations From....? __ to __			
Work Centre.....? _____			
OR			
Work Stations From....? _____ to _____			
Date Range From..... 8/12/95 to 8/12/95			
Items From.....? _____ to _____			
Suggested/Firm/Both.... 1 (1=Suggested 2=Firm,Blank=Both)			
Report Sequence..... 1 (1=WIP Location, 2=By Work Station)			
Include Subcontractors. 2 (0=No Subcontract, 1=Subcontract Only 2=Subcontract and Non-Subcontract)			
F3=Exit F4=Prompt F8=Submit job			
05-35 SF MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model code.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report. If left blank, there is no limit. Only WIP locations in the selected organisational model are included.

Work Centre

A work centre is a group of work stations, which may be specified for the report instead of a range of work stations. If neither is entered, then all work stations are included.

Work Stations From/To

This sets the limits for work stations to be included on the report if work centre is not specified.

If left blank, there is no limit. Only work stations that are in the selected organisational model are included.

Date Range From/To

These are mandatory fields. They set the date range for which the report is to be generated. The default range restricts the report to the current day but this can be extended. The range can be from one day up to thirty days.

Items From/To

A range of items can be entered. This may be used to restrict the report to a group of related items or an individual item.

Firm/Suggested/Both

Select which item schedules are to be included on the report. Suggested schedules are created by MRP and MPS by examining the supply and demand situation for items.

Firm schedules are suggested schedules that have been confirmed via the Confirm Suggested Schedules routines in MPS and MRP. These may be maintained.

The dates for which the schedules have been confirmed (and optionally maintained) is called the firm period.

If selected schedules are selected then they are included both inside and outside the firm period. Likewise this occurs if only firm schedules are selected. If both suggested and firmed schedules are selected, then only firm schedules are included during the firm period, but both firm and suggested schedules are shown outside that period.

Enter blank to accept both schedule types.

1 - To use suggested schedules only (this is the default).

2 - To use Firm schedules only.

Report Sequence

This parameter controls the sequence in which the report is printed.

1 - To sequence by WIP location. This is the default.

2 - To sequence by work station.

Within the selected group the report then sequences by item/operation, with production schedule quantities grouped together for each item/operation in the relevant time slots.

Include Subcontractor

This allows production schedule operations carried out by subcontractors to be included on the report.

0 - Exclude subcontract operations.

1 - To report only subcontract operations.

2 - To include all operations. This is the default.

Firm schedules have been created from suggested schedules. Suggested schedules are those created by the MRP based upon the supply and demand position of each item. Suggested schedules reflect what is needed, even if unrealistic.

Firm schedules typically reflect the plan to which the organisation has committed.

The process of firming schedules is performed in batch based upon a user defined set of dates which establish what is known as the Firm Period.

If suggested schedules are selected they are included both inside and outside the Firm Period. If firm schedules are selected then they are included in the Firm Period, and any schedules that the user has firmed and moved outside the Firm Period (via maintenance) is also included. If both planned and firmed schedules are selected, then firm schedules only are included during the Firm Period, but outside that period, both firm and suggested schedules are shown.

When all required fields are completed and components/outputs confirmed as required, the booking details window are displayed and updated after each booking entry in the session.



Press **F8=Submit Job** to validate field entries and submit job.